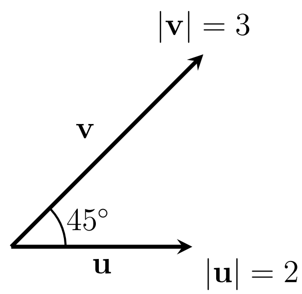
MTH 310 **Quiz 1** Name:____________________

Spring, 2026

Instructions. (15 points) Sections 12.1–12.4. Show relevant work for full credit.

(3pts) **1.** Use the figure below to answer the questions that follow. The vectors $\mathbf{u}$ and $\mathbf{v}$ lie in the plane of the page.



(a) Find $\mathbf{u} \cdot \mathbf{v}$.

Solution:

$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta = (2)(3)\cos 45^\circ = 6 \cdot \frac{\sqrt{2}}{2} = 3\sqrt{2}.$

(b) Find $|\mathbf{u} \times \mathbf{v}|$.

Solution:

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}| \sin\theta = (2)(3)\sin 45^\circ = 6 \cdot \frac{\sqrt{2}}{2} = 3\sqrt{2}.$

(c) Is $\mathbf{u} \times \mathbf{v}$ directed into the page or out of the page?

Solution:

By the right-hand rule, curling from $\mathbf{u}$ toward $\mathbf{v}$ (counterclockwise), $\mathbf{u} \times \mathbf{v}$ points **out of the page**.

(4pts) **2.** Find two unit vectors that are orthogonal to both $\mathbf{j} + 2\mathbf{k}$ and $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$.

Solution:

Let $\mathbf{a} = \langle 0, 1, 2\rangle$ and $\mathbf{b} = \langle 1, -2, 3\rangle$. Then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 1 & -2 & 3 \end{vmatrix} = \langle 7, 2, -1\rangle.$$

$|\mathbf{a} \times \mathbf{b}| = \sqrt{54} = 3\sqrt{6}$. The two unit vectors are $\pm \frac{1}{3\sqrt{6}}\langle 7, 2, -1\rangle$.

7 pts

(4pts) **3.** Find the vector projection of $\mathbf{b} = \langle 12, 1, 2\rangle$ onto $\mathbf{a} = \langle -1, 4, 8\rangle$.

Solution:

$\mathbf{a} \cdot \mathbf{b} = -12 + 4 + 16 = 8, \quad |\mathbf{a}|^2 = 1 + 16 + 64 = 81.$

$\text{proj}_{\mathbf{a}}\mathbf{b} = \dfrac{8}{81}\langle -1, 4, 8\rangle = \left\langle \dfrac{-8}{81}, \dfrac{32}{81}, \dfrac{64}{81} \right\rangle.$

(4pts) **4.** Find the angle at vertex B, to the nearest tenth of a degree, for the triangle with vertices $A(1, 0, -1)$, $B(3, -2, 0)$, and $C(1, 3, 3)$.

Solution:

$\overrightarrow{BA} = \langle -2, 2, -1\rangle, \quad \overrightarrow{BC} = \langle -2, 5, 3\rangle.$

$\overrightarrow{BA} \cdot \overrightarrow{BC} = 4 + 10 - 3 = 11.$

$|\overrightarrow{BA}| = 3, \quad |\overrightarrow{BC}| = \sqrt{38}.$

$\cos \angle B = \dfrac{11}{3\sqrt{38}}, \quad \angle B \approx 53.6°.$

8 pts